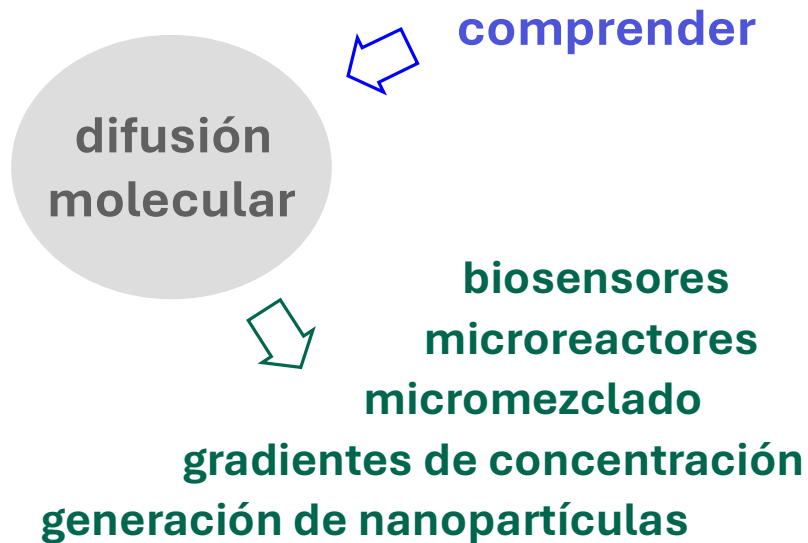


Índice del tema v

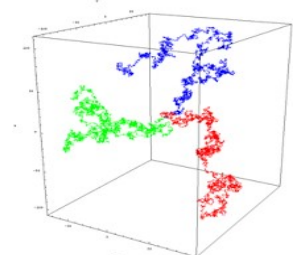
▶ v. Difusión y transporte

- ▶ Difusión molecular
- ▶ Transporte de especies
- ▶ Dispersión en coflujo
- ▶ Dispersión mecánica (medios porosos)
- ▶ La difusión como problema: micromezclado
- ▶ La difusión como solución: filtro-H y sensor-T
- ▶ Generación de gradientes de concentración

v. Problema central de los lab-on-a-chip



v. Difusión molecular



$$\sigma^2 = 6Dt$$
$$D = k_B T / 6\pi\mu r$$

v. Movimiento Browniano

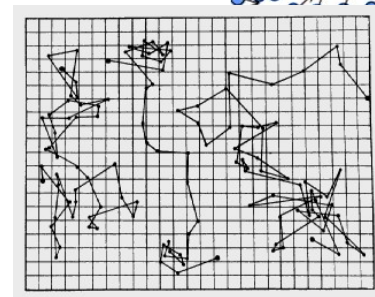
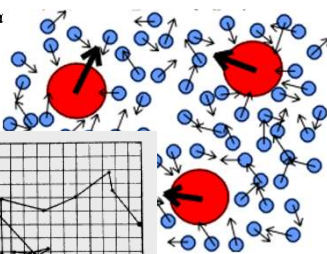
Ferner fanden wir in § 4 für den Mittelwert der Verschiebungen der Teilchen in Richtung der X -Achse in der Zeit t :

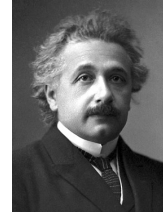
$$\lambda_x = \sqrt{2Dt}.$$

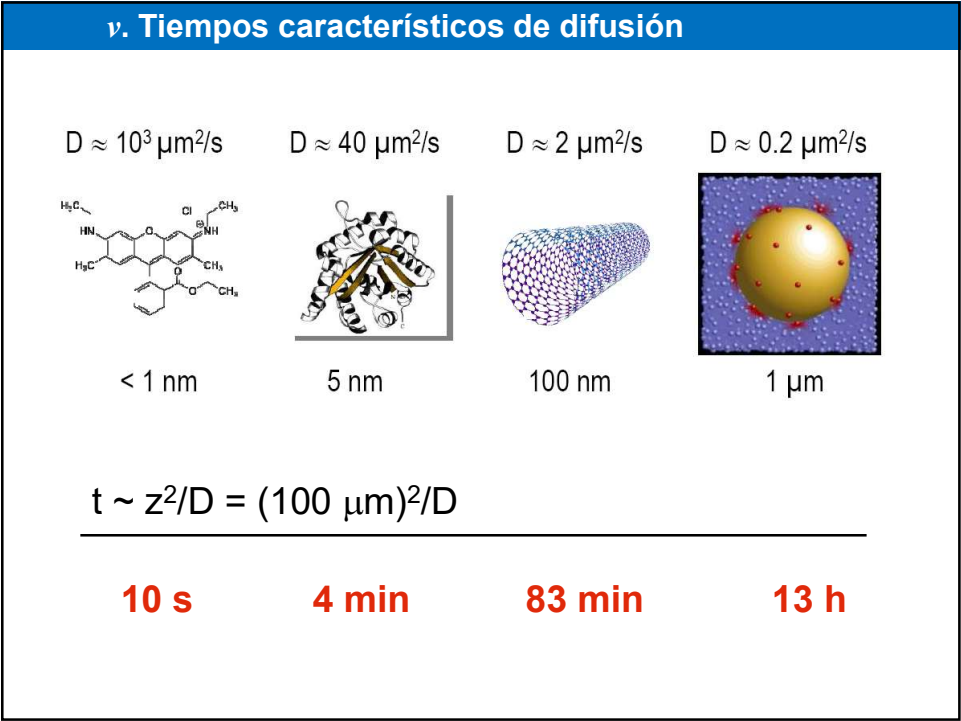
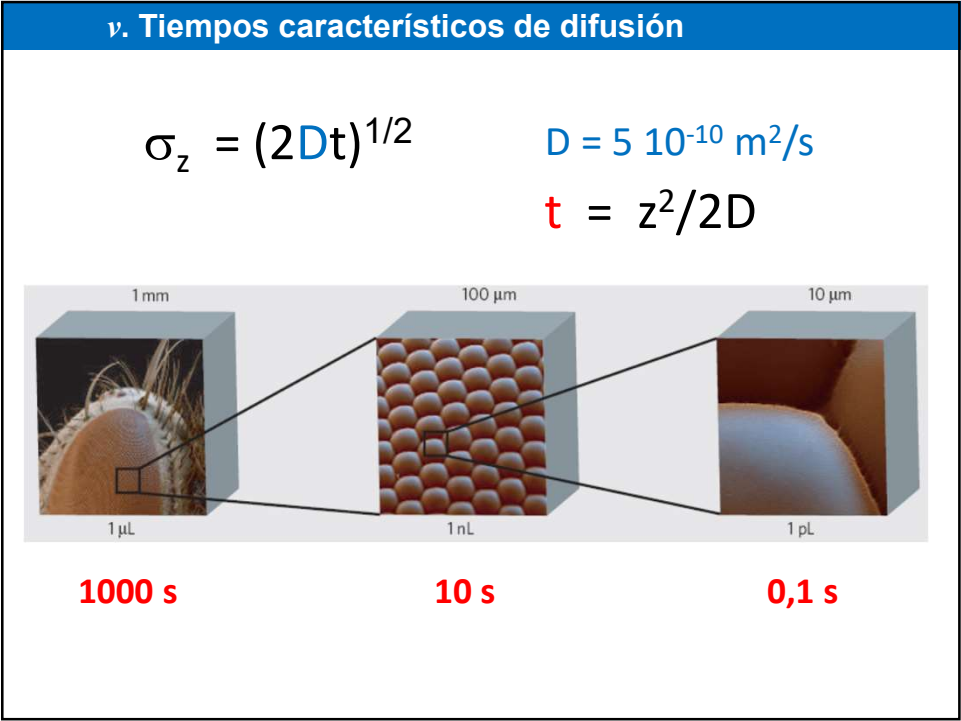
Durch Eliminieren von D erhalten wir:

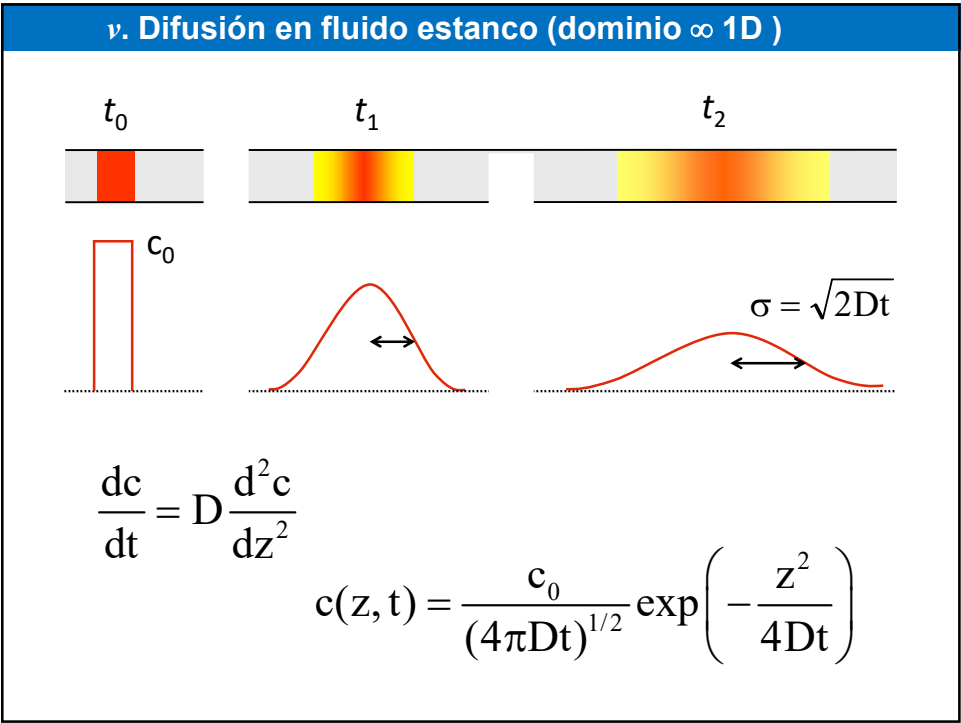
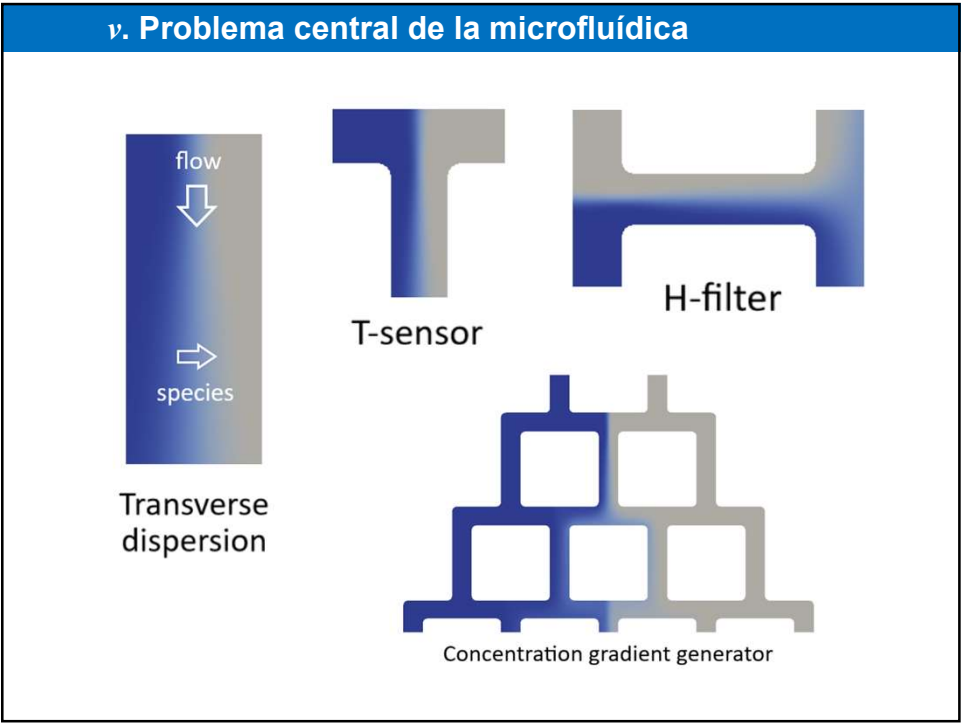
$$\lambda_x = \sqrt{t} \cdot \sqrt{\frac{RT}{N} \frac{1}{3\pi kP}}.$$

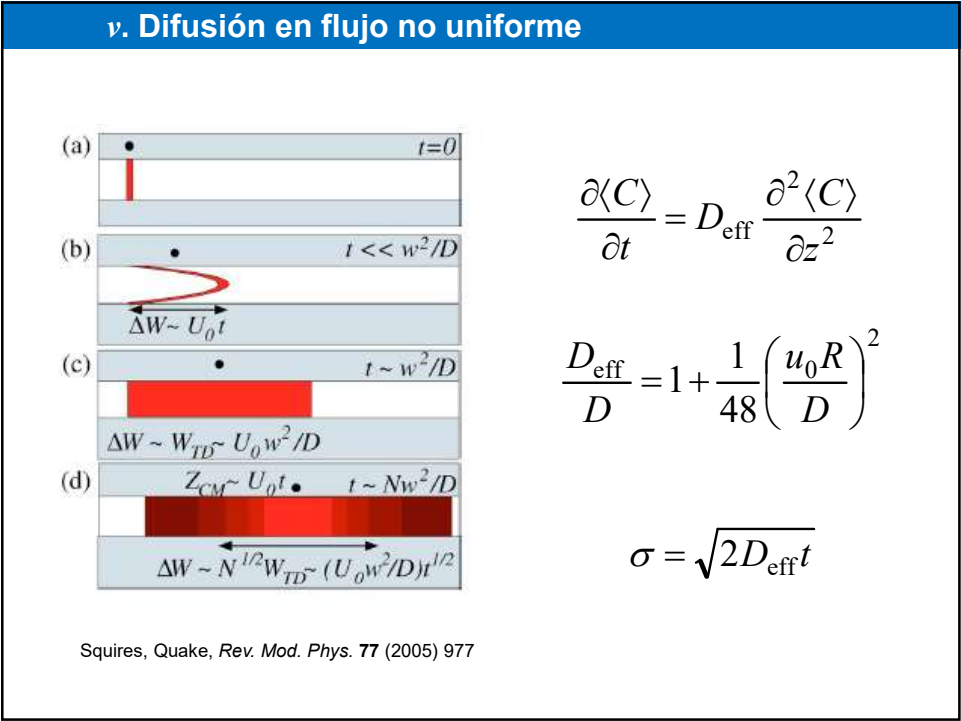
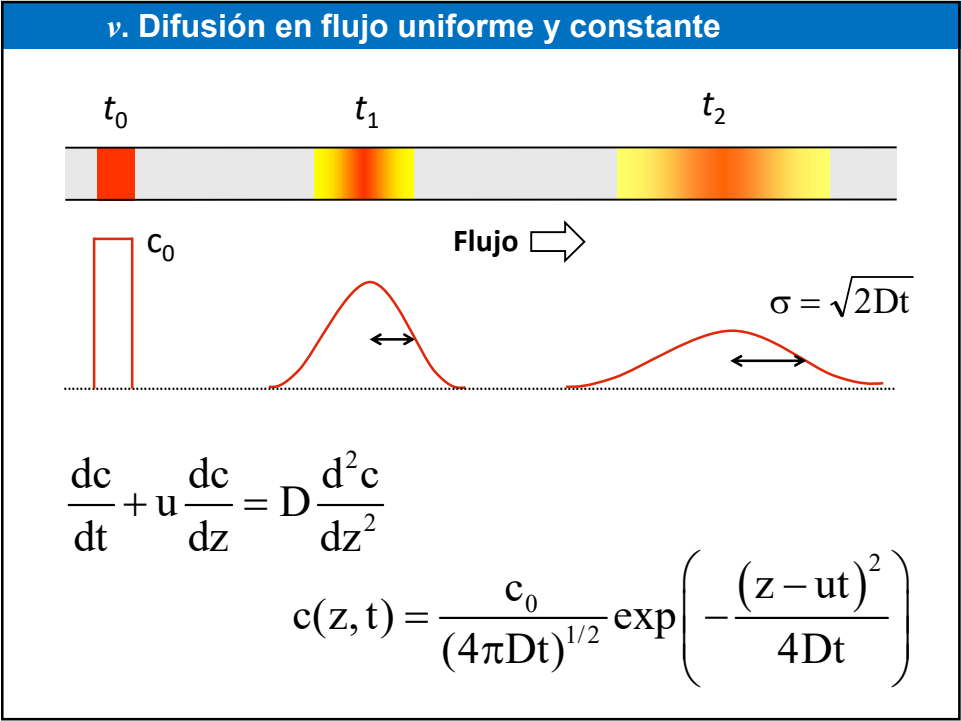
Diese Gleichung läßt erkennen, was die Faktoren hängen muß.

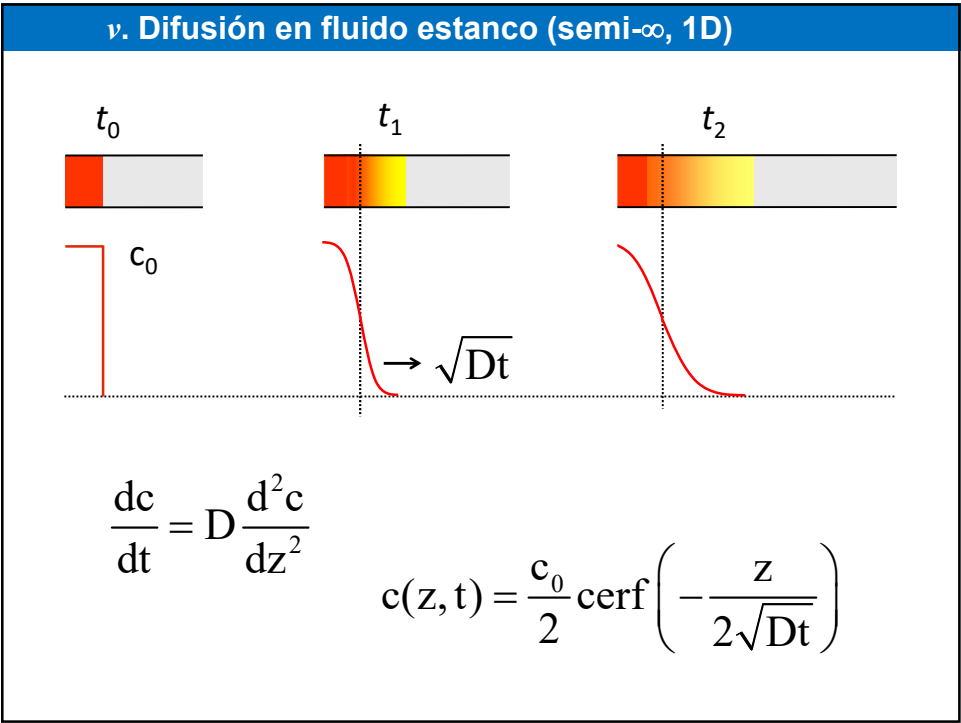
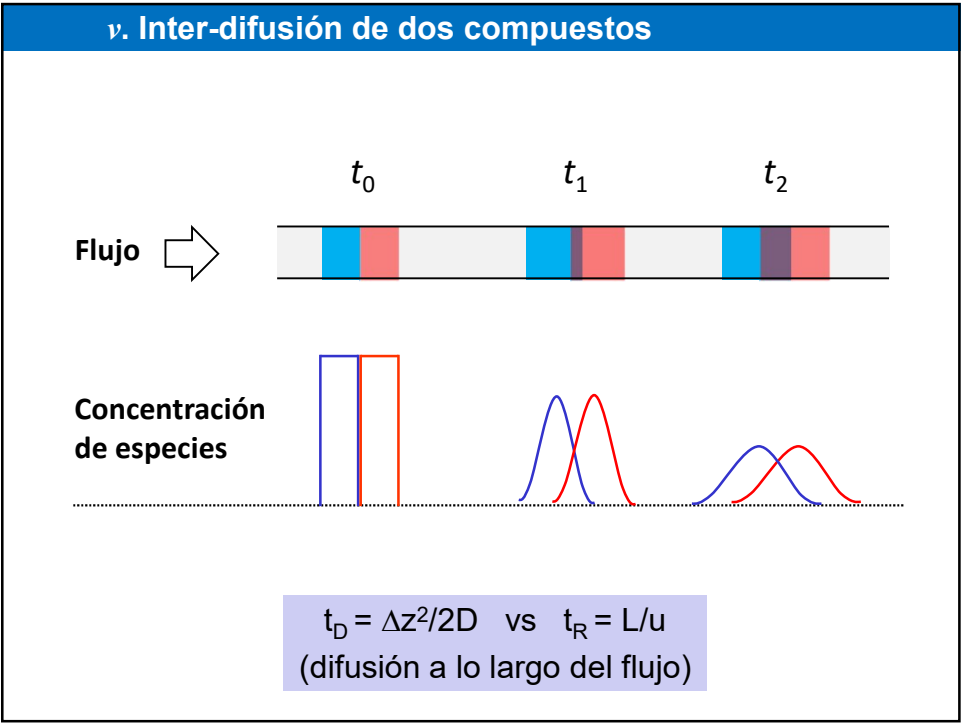


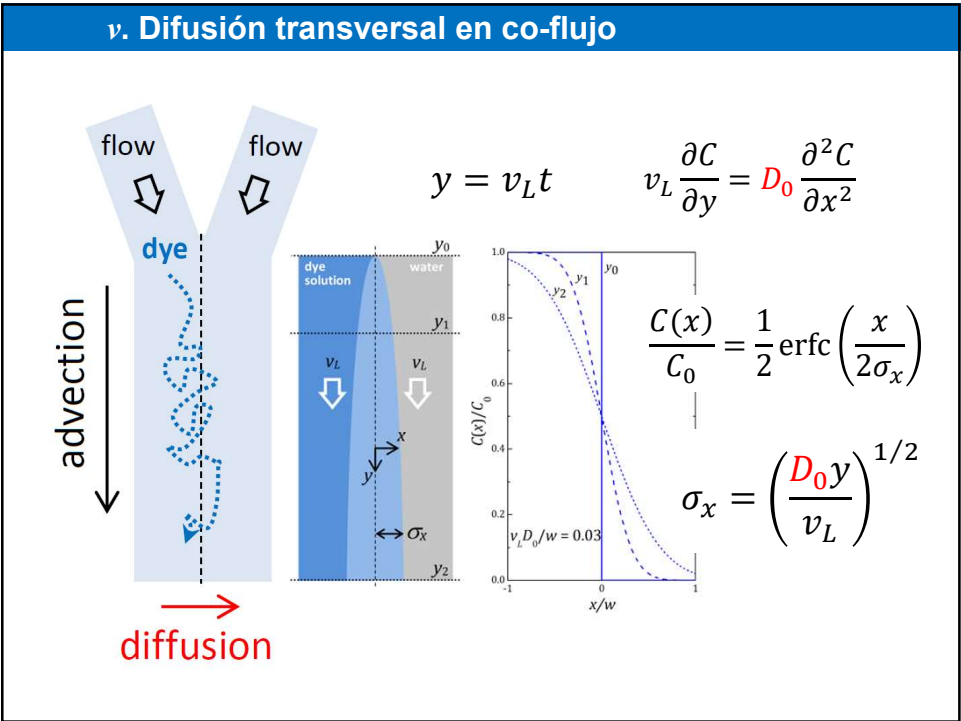
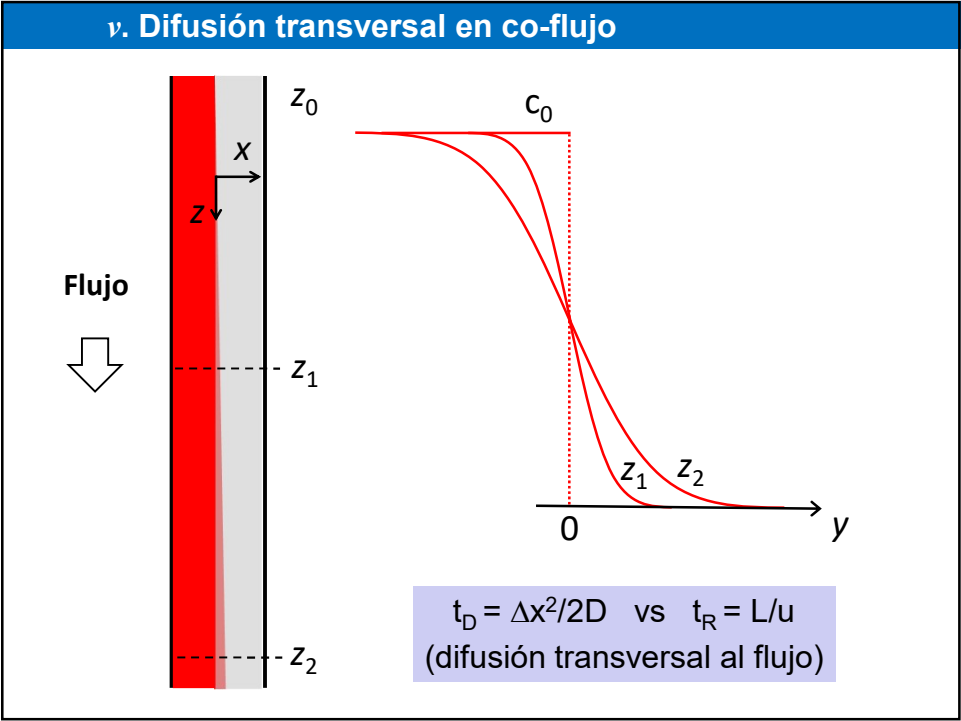








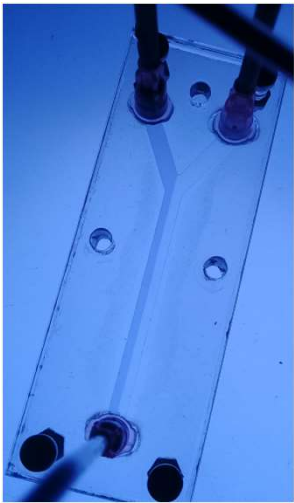




ν. Difusión transversal en co-flujo



T-sensor



Slit uchannel
250 μm x 4 mm



4 mL/h



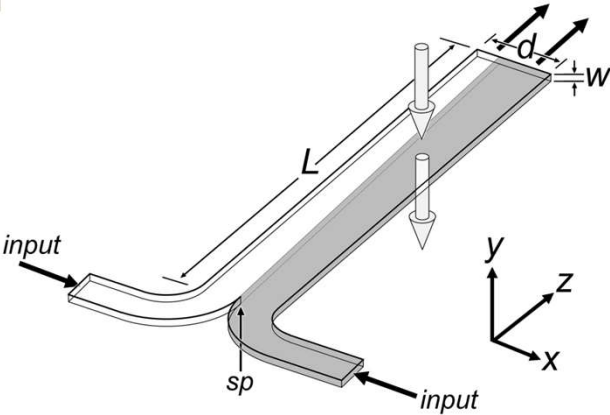
0.1 mL/h

Fotos: F. Minetti

ν. Difusión transversal en co-flujo



T-sensor



input

sp

L

d

w

input

y

z

x

Kamholtz et al., *Anal. Chem.* 71 (1999) 5340

v. La difusión como problema: micromezclado

$L = \frac{v w^2}{D}$

$w \sim 100 \mu\text{m}$

$L \sim \text{cm} - \text{m}$

$L \approx vt$

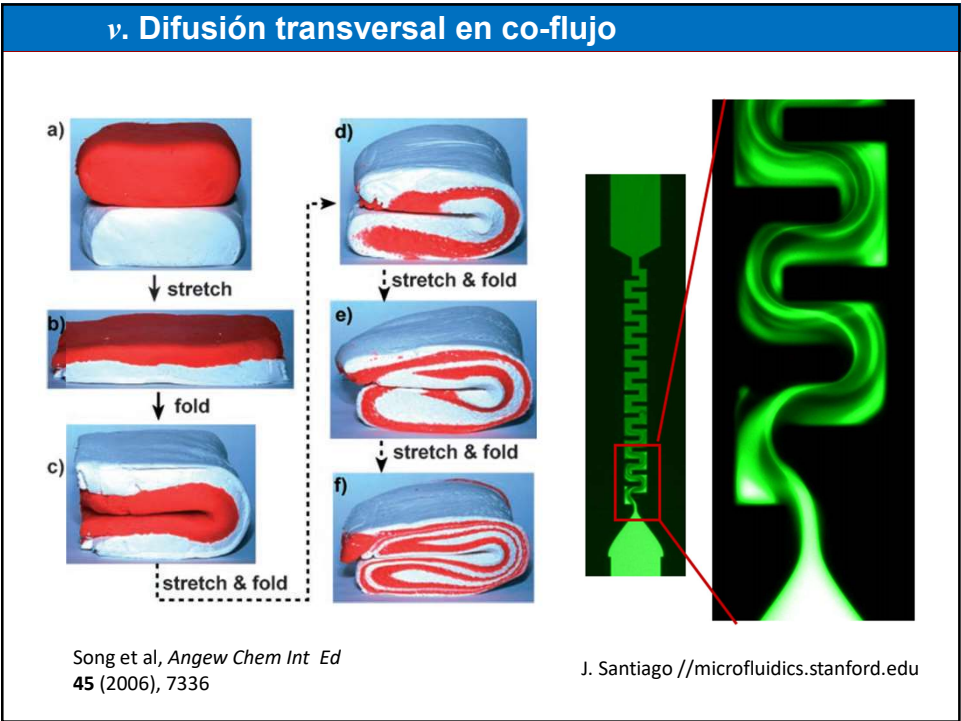
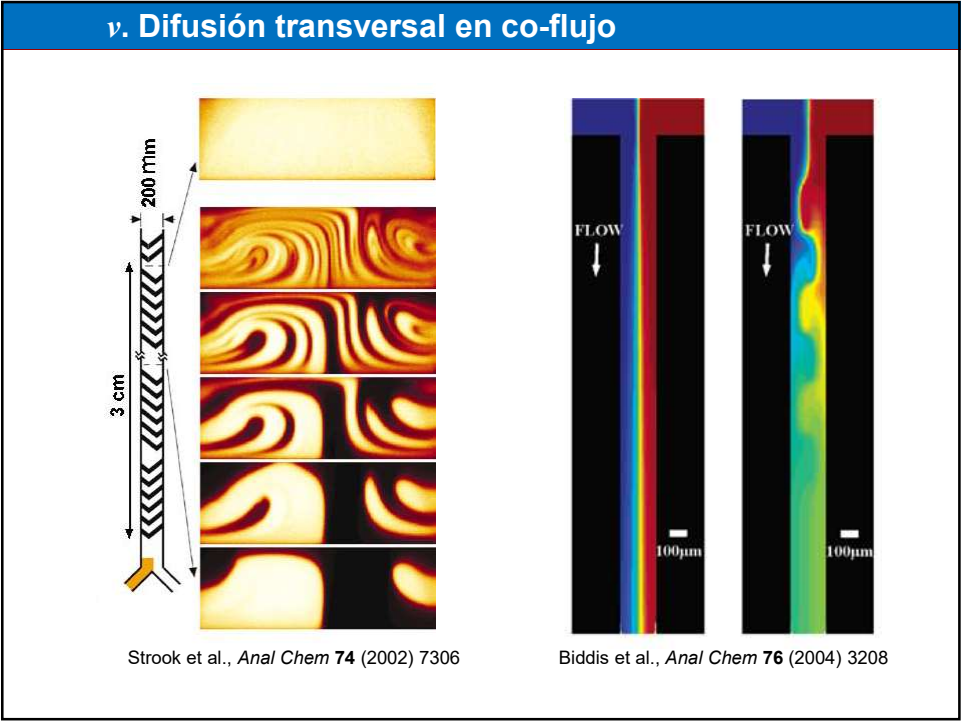
$w \approx (Dt)^{1/2}$

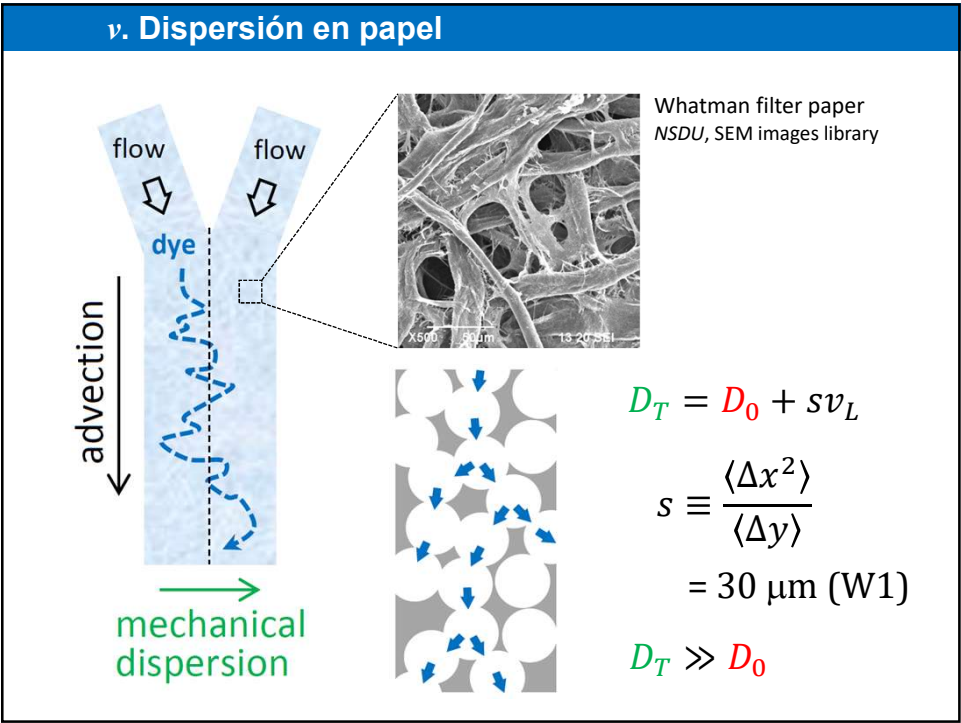
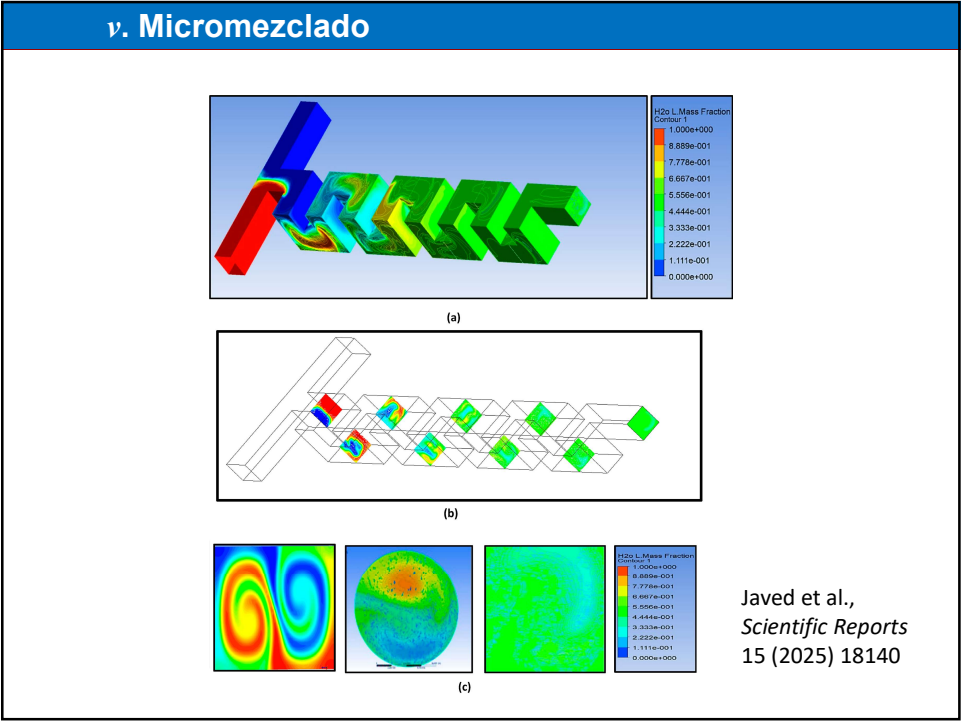
elveflow

v. La difusión como problema: micromezclado

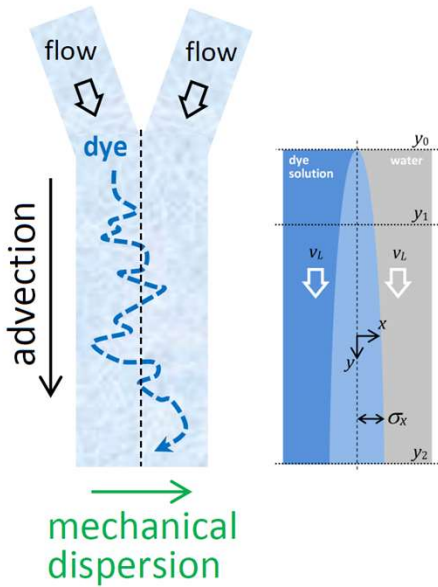
$L \approx vt$

$w \approx (Dt)^{1/2}$





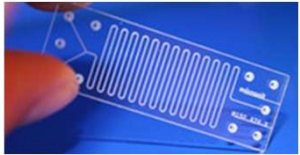
v. Dispersión en papel



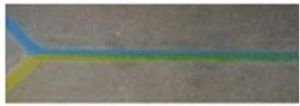
$$v_L \frac{\partial C}{\partial y} = D_T \frac{\partial^2 C}{\partial x^2}$$
$$\sigma_x = \left(\frac{D_T y}{v_L} \right)^{1/2}$$
$$D_T \approx s v_L$$
$$\sigma_x = (s y)^{1/2}$$

Urteaga et al, *Analyst* (2018) ...

v. Dispersión en papel

$$\sigma_{x,\mu ch} = \left(\frac{D_0 l_{\mu ch}}{v_L} \right)^{1/2}$$


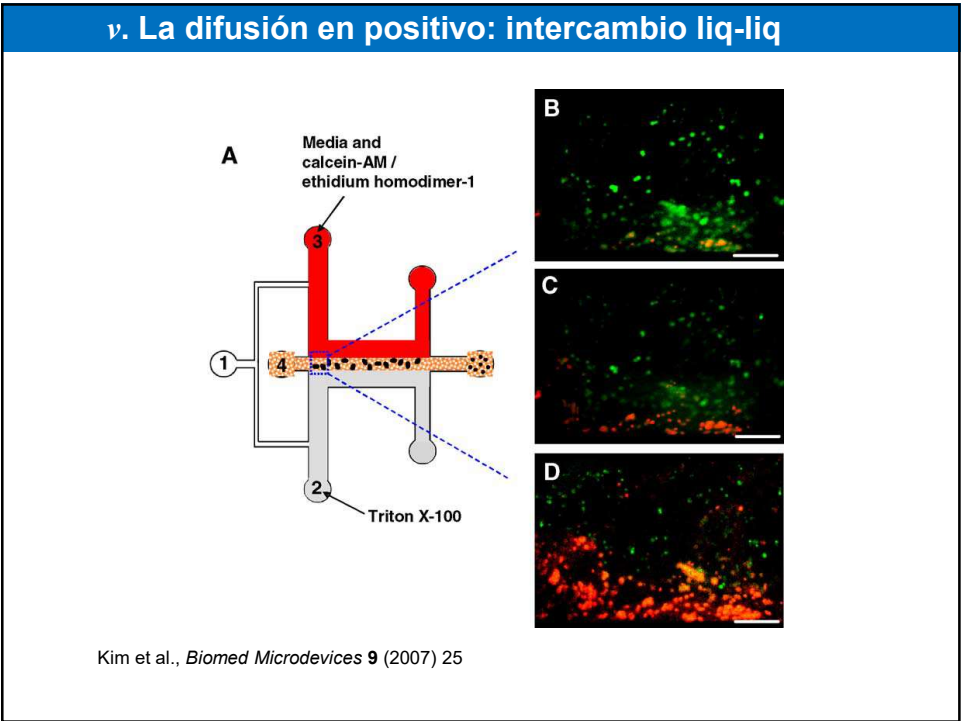
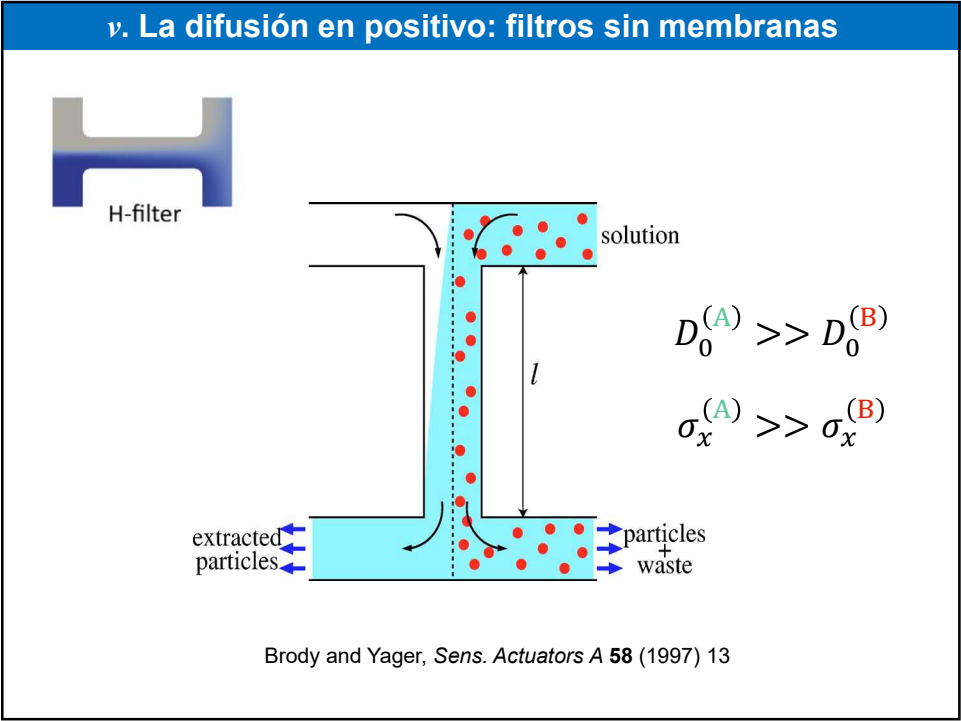
elveflow.com

$$\sigma_{x,paper} = (s l_{paper})^{1/2}$$


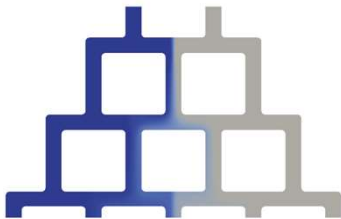
Resk et al, *Lab Chip*,
12 (2012) 773

$$\frac{l_{\mu ch}}{l_{paper}} = \frac{s v_L}{D_0} = \frac{D_T}{D_0}$$

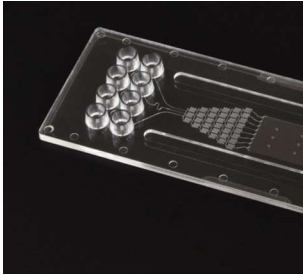
~ 10-100 ! (same dispersion width)

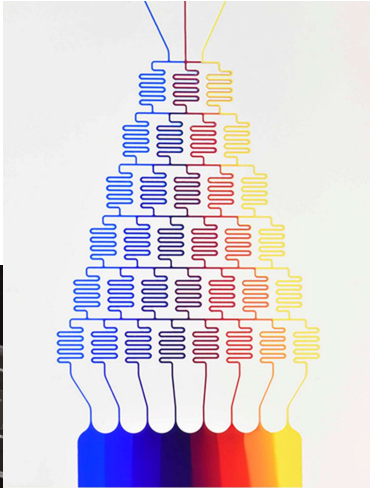


v. Generadores de gradientes de concentración



Concentration gradient generator

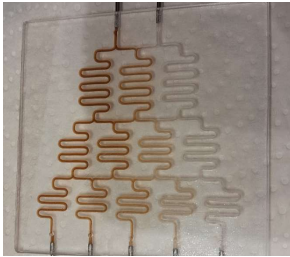


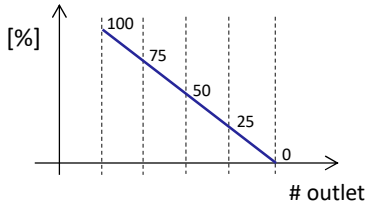


v. Generadores de gradientes de concentración

Solution A

Solvent

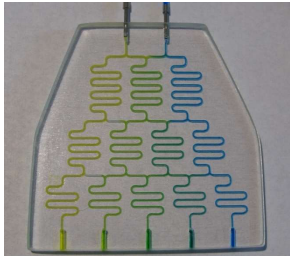


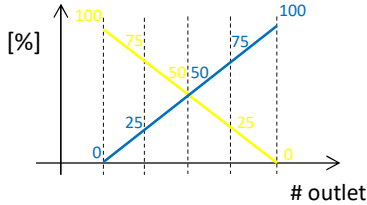


# outlet	Concentration [%]
1	100
2	75
3	50
4	25
5	0

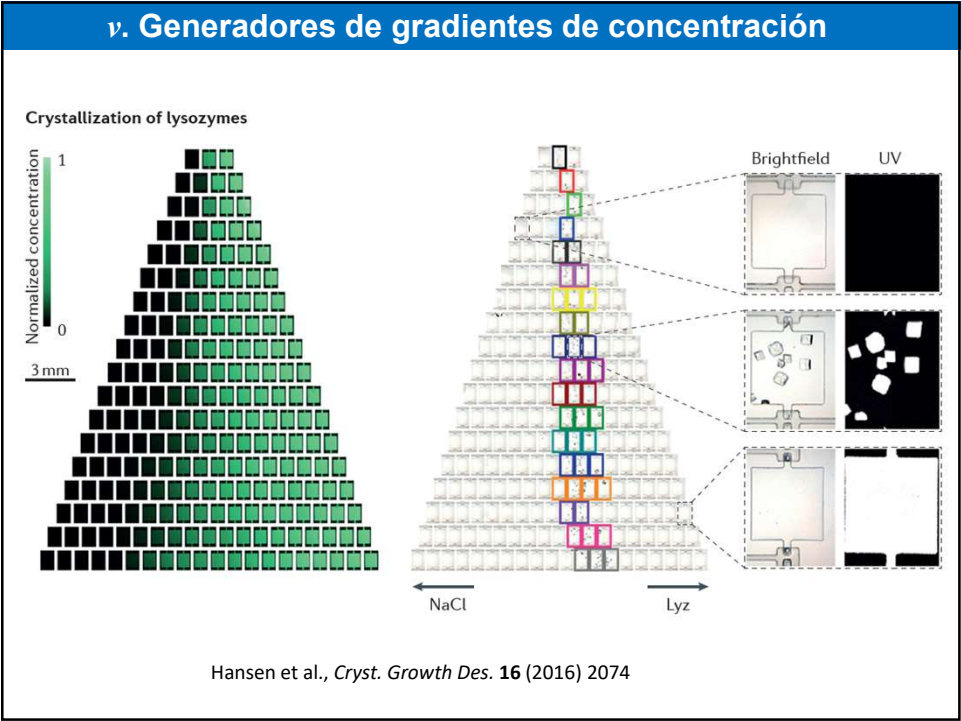
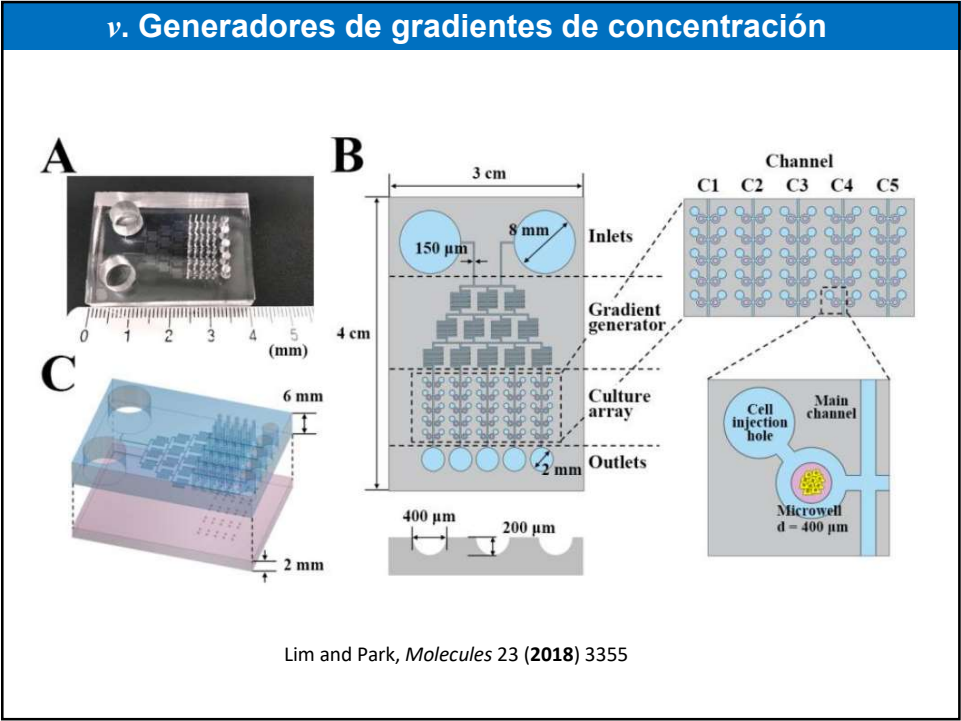
Solution B

Solution C





# outlet	Solution B [%]	Solution C [%]
1	100	0
2	75	25
3	50	50
4	25	75
5	0	100



v. Generadores de gradientes de concentración

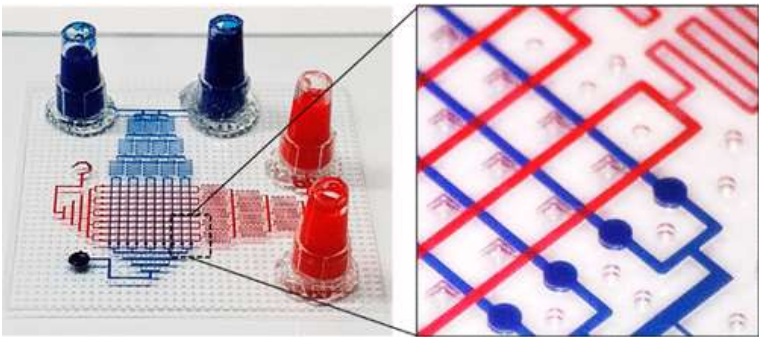
Generación de gradiente (dilutor) en escala logarítmica



Oh et al., *Lab Chip* **12** (2012) 515

v. Generadores de gradientes de concentración

Generación de gradientes en 2D: sistemas combinatoriales



Bae et al., *Anal Chem* **96** (2024) 5462